

Instructions: 1. Figures to the right indicate full marks for that question.
2. Use separate answer book for each section.
3. Assume suitable data whenever necessary and mention them clearly.

SECTION I

Q.1

(A) Choose correct answer from following multiple-choice questions. [07]

1. C++ Follows _____ approach
(a) Top down (b) Bottom up (c) Top-Bottom-Bottom (d) Bottom-Top
2. What the output of following code segment?
int ans=16%2; cout<<ans;
(a) 1 (b) 2 (c) 0 (d) -1
3. Default access specifier is _____
(a) public (b) default (c) private (d) protected
4. _____ is a special function, which is executed when object gets out of the scope.
(a) Destructor (b) Constructor (c) Inline function (d) Static function
5. _____ is a mechanism to give special meaning to the operator without changing its behavior.
(a) Type casting (b) Inheritance (c) Data Hiding (d) Operator Overloading
6. What the output of following code segment?
int ans=10 / 4.0; cout<<ans;
(a) 2.5 (b) 2.0 (c) 3.0 (d) 2.25
7. _____ functions are similar to macro.
(a) Inline (b) outline (c) in-out-in (d) baseline

(B) Answer the questions (Each question carries one mark) [04]

1. What is constructor?
2. _____ function can access private data member of a class and it is not a member function of a class.
3. What is class?
4. What is deep copy?

Q.2

(A) Explain about following Object Oriented Concepts: [06]

- (1) Encapsulation (2) Inheritance

(B) How to define member function in class? Explain with example. [06]

OR

Q.2

(A) What is variable? Explain with example. Write rules to define variable in C++. [06]

(B) Explain about following storage classes in C++. [06]

- (1) Auto (2) Extern (3) Static

Q.3

(A) Explain about following constructors with suitable example: [06]

(1) Parameterized Constructor (2) Default Constructor

(B) What is object? Explain with example. Explain characteristics of an object. [06]

OR

Q.3

(A) Explain about following objects with suitable example: [06]

(i) cin (ii) cout

(B) Explain about features of C++. [06]

SECTION II

Q.4

(A) Choose correct answer from following multiple-choice questions. [07]

1. Following data members are never inherited in derived class:

public (b) protected (c) private (d) mutable

2. What is the output of following code:

```
int a[10]={125,130,150,210,322}; cout<<*a+2;
```

(a) 127 (b) 130 (c) 123 (d) 150

3. _____ is a name given to a flow of data at the lowest level bytes

(a) Data (b) Stream (c) Sequence (d) Screen

4. What is the output of following code segment:

```
int x=5879; cout.width(8);
```

```
cout<<x;
```

(a)

(b)

(c)

(d)

				5	8	7	9
--	--	--	--	---	---	---	---

				5	8	7	9
--	--	--	--	---	---	---	---

				5	8	7	9
--	--	--	--	---	---	---	---

5	8	7	9
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5. The mechanism of deriving a new class from an old class is called_____

(a) Encapsulation (b) Polymorphism (c) Security (d) Inheritance

6. _____ flag is used to append data at the end of the file.

(a) ios::app (b) ios::bin (c) ios::in (c) ios::out

7. _____ class is used to read the data from the file.

(a) Stream (b) FILE (c) ifstream (d) ofstream

(B) Answer the questions (Each question carries one mark) [04]

1. What is abstract class?

2. What is method overloading?

3. What is exception?

4. Write the name of parent class from following code segment:

```
class A{};
```

```
class B:public A{};
```

Q.5

- (A) Explain about following concepts in details: [06]
(i) Single Inheritance (ii) Multilevel Inheritance
- (B) Differentiate between static (early) binding and dynamic (late) binding with suitable example. [06]

OR

Q.5

- (A) Explain about try/catch block with suitable example. [06]
- (B) Explain about virtual function. Write any two rules of virtual function. [06]

Q.6

- (A) Explain about getline() function in details. [06]
- (B) Explain about following file handling functions with suitable example: [06]
(i) read () (ii) write()

OR

Q.6

- (A) Explain about following file handling functions with suitable example: [06]
(i) seekg() (ii) tellg ()
- (B) Explain about following functions of formatted output with suitable example: [06]
(i) fill() (ii) precision()